

Adaptive Gait Generation and Postural Stabilization Framework for Quadruped Robots via IMU Feedback Integration

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Abstract—This paper presents a lightweight, computationally efficient motion control framework for articulated quadruped robots that achieves stable blind locomotion using exclusively proprioceptive inertial feedback. A strictly decoupled dual pipeline control architecture isolates rhythmic gait generation from reactive postural stabilization, enabling independent tuning and fault tolerant operation. Foot trajectories are shaped using quintic polynomial interpolation functions $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ that enforce zero velocity and zero acceleration at boundary conditions, eliminating mechanical impact transients. A multistage signal conditioning pipeline incorporating exponential moving average (EMA) prefiltering, configurable deadband suppression, gain scheduled PID with back calculation antiwindup, and cascaded lowpass filtering produces smooth corrective outputs free of high frequency jitter. During dynamic locomotion, phase aware rotational compensation matrices $R_y(\theta)R_x(\phi)$ are selectively applied to stance phase limbs via real time foot altitude thresholding, while swing phase limbs maintain unmodified trajectories to preserve stride clearance. Simulation based validation confirms the system's capacity to reject aggressive lateral disturbances while maintaining the Zero Moment Point (ZMP) within the support polygon.

I. INTRODUCTION

Legged robots possess the innate structural capacity to navigate discontinuous terrain and extreme topographies that inherently challenge conventional wheeled platforms. However, developing resilient locomotion policies capable of stabilizing a heavily articulated system with multiple degrees of freedom against unmodeled disturbances remains a rigorous control challenge.

Contemporary solutions rely heavily on computationally intensive algorithms, such as Model Predictive Control (MPC) and Reinforcement Learning (RL), to estimate optimal contact dynamics [1], [2]. While highly performant, these stochastic solvers predominantly require specialized hardware accelerators or processors operating at high frequencies, severely precluding deployment upon the embedded edge devices ubiquitous in modern mechatronics.

Conversely, Central Pattern Generators (CPGs) driven by heuristic rules and paired with reactive reflex feedback loops can approximate biological stability at a fraction of the computational cost. When inertial telemetry of high fidelity, specifically instantaneous roll and pitch orientation [3], is directly integrated into kinematic calculations operating in real time, robust blind ambulation becomes tractable. For instance, recent studies have demonstrated the use of posture stabilization algorithms to maintain a level torso when mapping static gaits

onto swaying and oscillating platforms without visual sensors [4]. Inspired by these stability principles, this work proposes four key contributions that distinguish it from conventional CPG-based quadruped controllers:

- 1) A **quintic polynomial trajectory planner** utilizing the normalized interpolation function $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ to enforce zero velocity and zero acceleration boundary conditions at both liftoff and touchdown, eliminating mechanical impact transients that corrupt inertial sensor baselines.
- 2) A **multistage signal conditioning pipeline** comprising: (a) exponential moving average (EMA) prefiltering, (b) configurable deadband suppression, (c) gain scheduled PID with integral leakage and back calculation antiwindup [5], and (d) cascaded lowpass filtering, producing smooth corrective outputs free of servo jitter.
- 3) A **phase-aware stance correction** mechanism that applies rotational compensation matrices $R_y(\theta)R_x(\phi)$ exclusively to limbs classified as load bearing via foot altitude thresholding in real time ($|Z - Z_{stance}| \leq Z_{thresh}$), while swing phase limbs maintain unmodified trajectories.
- 4) A **strictly decoupled dual-node architecture** that isolates rhythmic gait generation from reactive postural stabilization, enabling independent tuning, independent failure modes, and asynchronous execution at differing control frequencies.

The remainder of this paper is organized as follows: Section II details the distributed hardware and software architecture. Section III formulates the inverse kinematics and polynomial trajectory framework. Section IV outlines the adaptive PID balancing logic. Section V validates the framework with quantitative diagnostic results, and Section VI presents conclusions and pathways for future research.

II. SYSTEM ARCHITECTURE

A. Node Topology

The framework operates as a distributed processing architecture implemented natively in ROS 2, horizontally partitioned across two functionally isolated computational nodes. The primary *Gait Generator* node manages all rhythmic trajectory computation, inverse kinematics resolution, and actuator command dispatch over the serial bus. Concurrently, an asynchronous *Balance Controller* node exclusively handles high-frequency inertial measurement acquisition, multistage signal conditioning, and PID correction computation.

Inter-process communication between the two nodes leverages the ROS 2 Data Distribution Service (DDS) via a strictly decoupled publish-subscribe structure. The balance node streams lightweight corrective vector messages across dedicated asynchronous topics. This architectural segregation guarantees that trajectory generation processing bottlenecks cannot preempt or block critical sensory compensations, enabling deterministic, fault-tolerant operation even under severe CPU load.

B. State Representation

Global governance overlying the control architecture is dictated by boolean gating, handling two discrete operating states:

- **HOME:** A static configuration lacking cyclic translation vectors. The controller computes uniform adjustments in joint space rigidly mapped to a resting stance. Base angular deviations translate proportionally into antagonistic torque compensations aimed at continuous postural rectification. A configurable startup delay of T_{delay} seconds prevents premature correction during system initialization transients.
- **GAIT:** The dynamic state active during commanded planar translations. The platform executes cyclic polynomial trajectories using configurable phase shift arrays to generate trot ($\phi = [0, 0, 0.5, 0.5]$), walk ($\phi = [0, 0.25, 0.5, 0.75]$), and other gaits involving multiple limbs. Spatial corrections derived from inertial feedback are applied exclusively to limbs in stance phase, while swing legs maintain zero compensation to ensure uninhibited stride clearance.

III. KINEMATICS AND GAIT PLANNING

A. Inverse Kinematics

To mandate the endpoint position (x, y, z) of each limb relative to the geometric origin of the base frame, a nonlinear inverse kinematics (IK) model [6] derives the required articular angles $(\theta_1, \theta_2, \theta_3)$ for the hip, thigh, and calf joints. The physical topology structuring the Cartesian workspace includes a platform length of $L = 250$ mm, a platform width of $W = 193$ mm, and individual linkage spans $l_1 = 45$ mm, $l_2 = 107$ mm, and $l_3 = 116$ mm.

The mathematical solution avoids geometric singularities by enforcing strictly restricted bounding limits calculated dynamically during system initialization. Fig. 1 shows the physical quadruped robot prototype alongside its simulation model, which illustrates the reference frame assignments at each joint origin, with RGB axes denoting the local coordinate systems used for forward and inverse kinematic resolution. The kinematic chain is mathematically formalized using the Modified Denavit-Hartenberg (MD-H) convention [6], meaningfully derived from the exact coordinate transformations in the software base. The precise parameter set defining the physical mechanism is presented in Table I, capturing the structural distinctions between the left and right limbs.

The inverse kinematics of the physical model establish the relationship between the local target coordinate (x, y, z) and the articular joint space. Due to mirrored coordinate reflections, the

TABLE I: Modified Denavit-Hartenberg (MD-H) Parameters

Link (i)	α_{i-1}	a_{i-1}	d_i		θ_i
			Left Limbs	Right Limbs	
1 (Coxa)	0°	0	0	0	θ_1^*
2 (Femur)	90°	0	$-l_1$	l_1	θ_2^*
3 (Tibia)	0°	l_2	0	0	θ_3^*
4 (Foot)	0°	l_3	0	0	0

* Variable articulation angle

mechanical system equations decouple and the corresponding joint angles can be solved as follow:

$$\begin{cases} \theta_1 = k \left[\text{atan2}(-x, y) + \text{atan2}(-\gamma, k l_1) \right] \\ \theta_2 = \text{atan2}(p_2, p_1) - \text{atan2}(l_3 s_3, l_2 + l_3 c_3) \\ \theta_3 = \text{atan2}(s_3, c_3) \end{cases} \quad (1)$$

where $k \in \{-1, 1\}$ is the structural symmetry coefficient ($k = -1$ for Left Limbs, $k = 1$ for Right Limbs), and the geometric intermediates are derived symmetrically as:

$$\begin{aligned} \gamma &= \sqrt{x^2 + y^2 - l_1^2} \\ p_1 &= x \cos \theta_1 + y \sin \theta_1 \\ p_2 &= -k z \\ c_3 &= \frac{p_1^2 + p_2^2 - l_2^2 - l_3^2}{2 l_2 l_3} \\ s_3 &= k \sqrt{1 - c_3^2} \end{aligned}$$

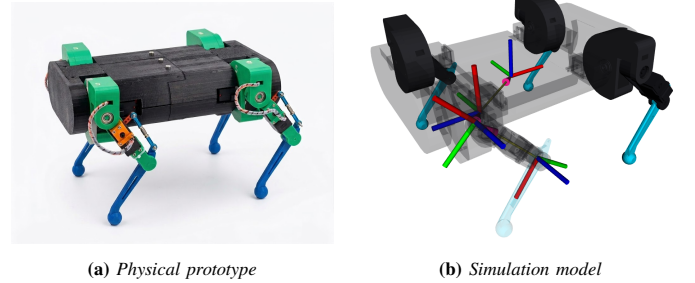


Fig. 1: Quadruped robot physical prototype and simulation model with reference frames

B. Quintic Trajectory Formulation

Minimizing ground impact velocity is mathematically imperative; violent mechanical shocks corrupt inertial sensor baselines and induce structural fatigue [7]. Therefore, foot trajectories are shaped using the normalized quintic interpolation function:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5 \quad (2)$$

where $\tau = t/T \in [0, 1]$ and $s \in [0, 1]$. This function enforces the critical boundary conditions:

$$\begin{aligned} s(0) &= 0, & s(1) &= 1 \\ \dot{s}(0) &= \dot{s}(1) = 0 \\ \ddot{s}(0) &= \ddot{s}(1) = 0 \end{aligned} \quad (3)$$

The resulting velocity and acceleration profiles are obtained by differentiation:

$$\dot{s}(\tau) = \frac{1}{T}(30\tau^2 - 60\tau^3 + 30\tau^4) \quad (4)$$

$$\ddot{s}(\tau) = \frac{1}{T^2}(60\tau - 180\tau^2 + 120\tau^3) \quad (5)$$

Fig. 3a visualizes the position, velocity, and acceleration profiles, confirming the enforced zero derivative boundary conditions.

For the swing phase of duration $T_{swing} = 70$ frames, the foot transitions from liftoff coordinate **A** to touchdown coordinate **D** along a D shaped parabolic arc with configurable lift height $h_{lift} = 40$ mm:

$$\mathbf{p}_{swing}(\tau) = \mathbf{A} + s(\tau)(\mathbf{D} - \mathbf{A}) \quad (6)$$

$$z(\tau) = z_{ground} + 4h_{lift} \cdot s(\tau)(1 - s(\tau)) \quad (7)$$

The resulting D shaped foot path is illustrated in Fig. 2, showing the swing arc and ground retrace phases with annotated liftoff and touchdown boundaries.

3D Foot Trajectory (Left Front)

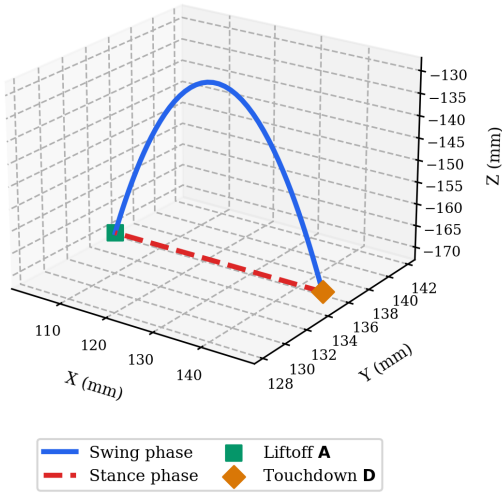
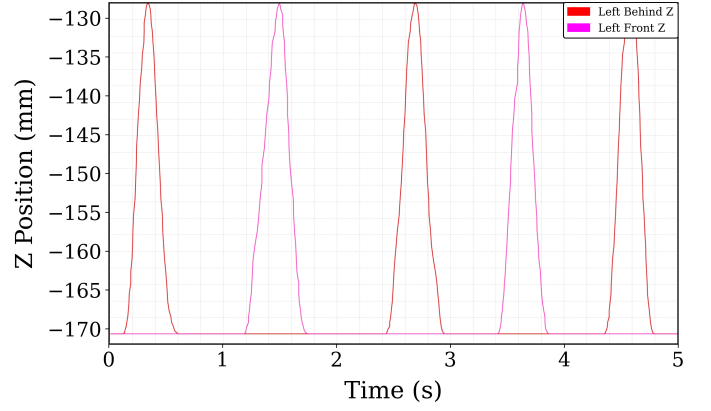


Fig. 2: 3D foot trajectory for the left front limb

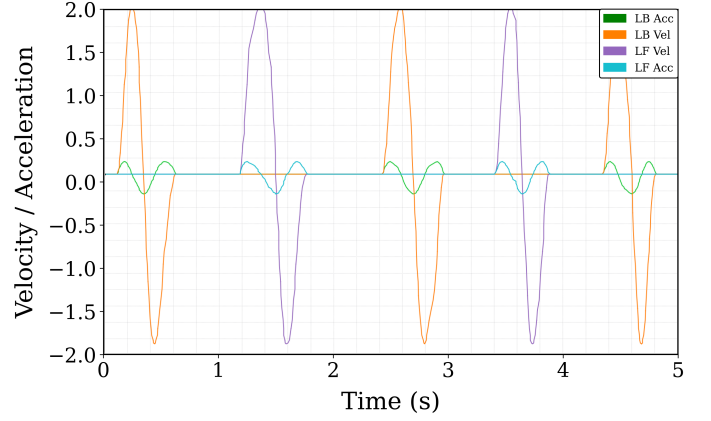
During the stance phase of duration $T_{stance} = 200$ frames, the foot retraces along the ground plane using the same quintic interpolation to ensure smooth velocity profiles in the reverse direction. The stride length is set at $L_{stride} = 45$ mm, centered about each leg's neutral x_{center} position.

Overlapping gaits involving multiple limbs are generated by applying phase shift offsets ϕ_i to the trajectory index for each limb. For a standard trot, diagonal pairs share identical phase offsets $\phi = [0.0, 0.0, 0.5, 0.5]$, producing the characteristic diagonal synchronization with two beats.

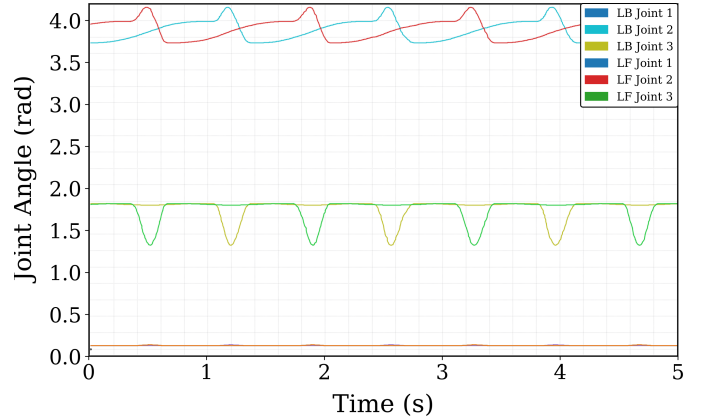
Fig. 3b presents the resolved joint angles $\theta_1, \theta_2, \theta_3$ for the left front limb over one complete gait cycle, with swing and stance phases explicitly shaded.



(a) Cartesian trajectories



(b) Velocities and accelerations



(c) Resolved joint angles

Fig. 3: Left front limb gait trajectory mapping

IV. IMU BASED POSTURAL STABILIZATION

Inertial corrections are computed asynchronously within the strictly isolated Balance Controller. Raw quaternion orientation data is converted to Euler roll (ϕ) and pitch (θ) angles [3] and processed through a multistage transformation pipeline.

A. Static Home Posture Correction

During the static home phase, all four limbs maintain sustained ground contact. The correction pipeline proceeds in five sequential stages:

Stage 1: EMA Prefiltering. Raw inertial measurements are smoothed via an exponential moving average of the first order to suppress sensor noise at high frequencies:

$$\begin{aligned}\hat{m}_k &= \alpha_{raw} \hat{m}_{k-1} + (1 - \alpha_{raw}) m_k \\ \alpha_{raw} &= 0.4\end{aligned}\quad (8)$$

Stage 2: Deadband Suppression. A configurable deadband of ± 0.02 rad eliminates trivial oscillations that would otherwise generate unnecessary servo activity:

$$e_{dz} = \begin{cases} e - d & e > d \\ 0 & |e| \leq d \\ e + d & e < -d \end{cases}, \quad d = 0.02 \quad (9)$$

Stage 3: Gain Scheduled PID with Antiwindup. A full PID controller [5] with three novel enhancements processes the deadband filtered error:

$$\begin{aligned}u_{PID} &= K_P \cdot g_s \cdot e \\ &+ K_I \cdot g_s \cdot \mathcal{I} + K_D \cdot \dot{e}_{lpf}\end{aligned}\quad (10)$$

where the gain schedule factor $g_s \in [0.5, 1.0]$ scales proportionally to the error magnitude, preventing corrections that are overly aggressive near zero:

$$g_s = \text{clamp}\left(\frac{|e|}{e_{thresh}}, 0.5, 1.0\right) \quad (11)$$

where $e_{thresh} = 0.08$ rad.

The integral accumulator features exponential leakage decay ($\lambda = 0.9995$) to prevent windup at steady state, bounded by symmetric saturation limits ($|\mathcal{I}| \leq 1.5$). Additionally, back calculation antiwindup reduces the accumulator when the PID output exceeds saturation bounds:

$$\mathcal{I}_k = \mathcal{I}_k - \frac{u_{raw} - u_{sat}}{2K_I} \quad \text{if } |u_{raw}| > u_{max} \quad (12)$$

The derivative term is presmoothed with a dedicated LPF ($\alpha_d = 0.7$) and includes a derivative ramp threshold ($e_{ramp} = 0.005$) that zeroes the derivative when the error is negligibly small, preventing derivative kick.

Stage 4: Output Saturation. The composite PID output is clamped to $|u| \leq 1.5$ rad to protect actuator limits.

Stage 5: Output Lowpass Filtering. A final lowpass filter of the first order smooths the corrective output to eliminate residual step discontinuities:

$$\begin{aligned}H_k &= \alpha_c H_{k-1} + (1 - \alpha_c) u_{sat} \\ \alpha_c &= 0.15\end{aligned}\quad (13)$$

B. Phase Aware Dynamic Gait Correction

While the robot is actively ambulating, the correction pipeline switches to an adaptive PID controller specific to the gait cycle that incorporates three additional mechanisms:

Moving Average Baseline. IMU measurements are accumulated into a buffer spanning one full gait cycle. The running mean provides a smoothed estimate of the true tilt over the cycle, filtering out periodic oscillations inherent to legged locomotion.

Adaptive Proportional Gain. The proportional gain scales dynamically with the instantaneous tilt magnitude to provide stronger correction authority during large disturbances:

$$K_{P,eff} = K_P \left(1 + \min\left(1.0, \frac{\max(|\phi|, |\theta|)}{K_{prop}}\right) \right) \quad (14)$$

Zero Crossing Integral Reset. The integral accumulator is immediately reset to zero whenever the error signal crosses zero, preventing oscillatory overshoot during transient recovery.

Stance Phase Rotational Compensation. The corrective output is applied as a rotation matrix in the body frame $R = R_y(\theta)R_x(\phi)$:

$$R = \begin{bmatrix} c_\theta & s_\theta s_\phi & c_\phi s_\theta \\ 0 & c_\phi & -s_\phi \\ -s_\theta & s_\theta c_\phi & c_\phi c_\theta \end{bmatrix} \quad (15)$$

This matrix is applied exclusively to limbs satisfying the stance condition $|Z_{foot} - Z_{stance}| \leq Z_{thresh}$, where $Z_{stance} = -170$ mm and $Z_{thresh} = 5$ mm. Only the vertical component of the transformed foot position is modified; horizontal coordinates are preserved to maintain stride geometry:

$$\begin{aligned}\mathbf{p}_{adj} &= R \cdot \mathbf{p}_{raw} \\ x_{adj} &= x_{raw}, \quad y_{adj} = y_{raw}\end{aligned}\quad (16)$$

V. RESULTS AND DIAGNOSTIC DATA

A comprehensive diagnostic pipeline continuously broadcasts internal controller states across a structured topic hierarchy to objectively quantify algorithmic performance. Data is presented separately for the static home posture and dynamic trotting gait.

A. Static Home Posture Stabilization

During the static home posture, the quadruped maintains all four limbs in ground contact while the PID controller actively rejects base orientation disturbances.

Fig. 5 presents the pitch and roll error tracking response. The robust orientation error is shown to converge rapidly, settling completely within the configured ± 0.02 rad deadband threshold in approximately 1.5 seconds post-disturbance. This empirical convergence confirms that the deadband and integral anti-windup mechanisms effectively stabilize the system. The multistage pipeline produces appropriate, critically-damped actuator commands suitable for sustained balancing without triggering unwanted limit-cycle oscillations around the spatial setpoint.

B. Dynamic Trotting Gait Stabilization

During active trotting locomotion, the adaptive PID controller maintains postural stability while the quadruped executes cyclic quintic trajectories.

Fig. 6 displays the dynamic pitch and roll error during the trotting phase. Despite the periodic ground impact forces inherently generating transient orientation spikes of up to ~ 0.15 rad, the adaptive mechanisms correctly extract the sustained tilt trend from the severe cyclic gait artifacts. The phase-aware PID controller successfully binds the peak steady-state tracking

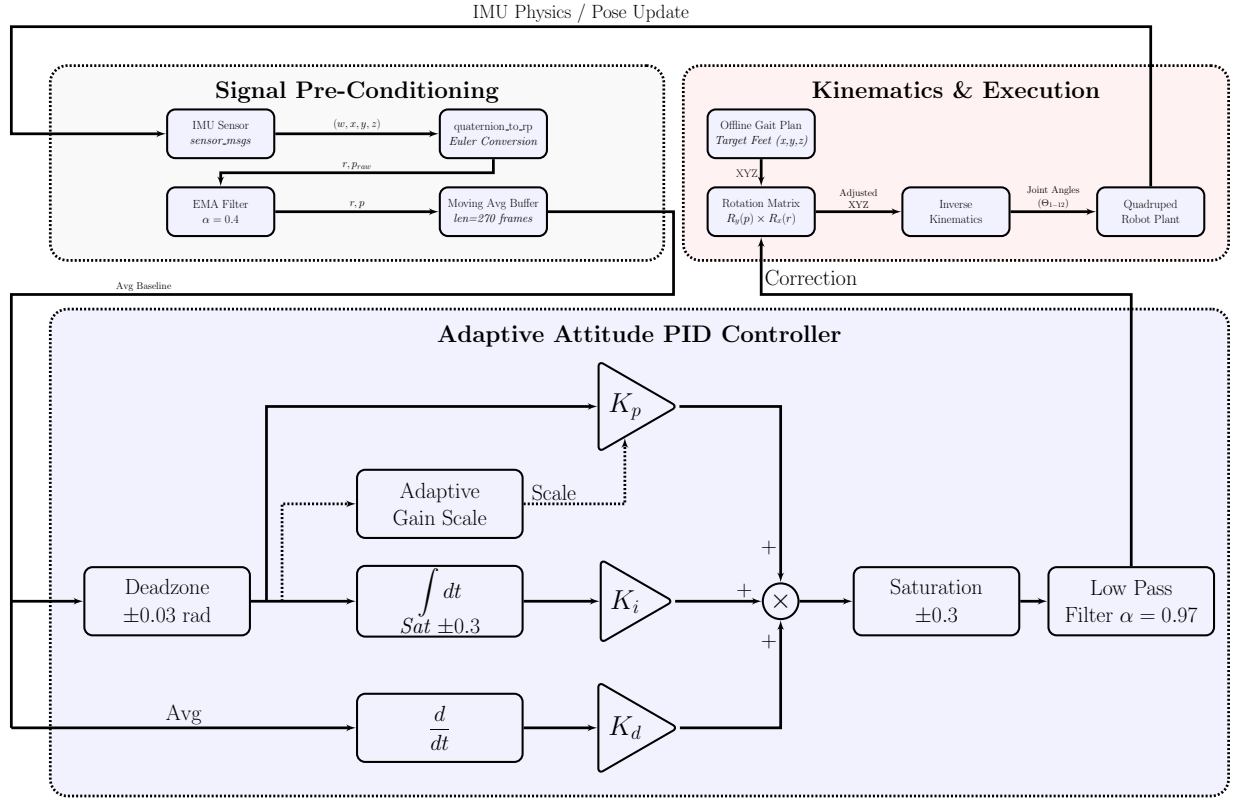


Fig. 4: Adaptive Attitude Adjustment PID Control Architecture

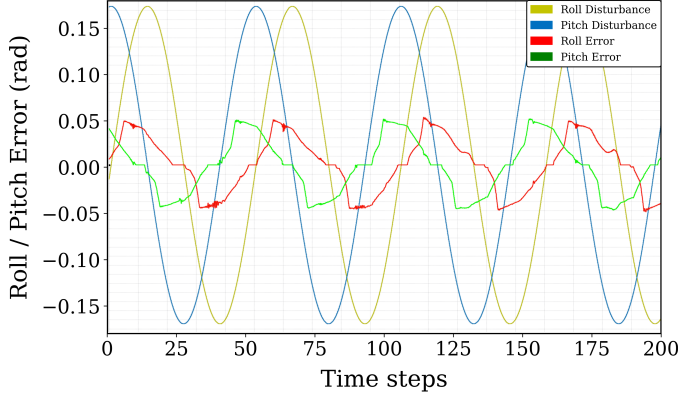


Fig. 5: Controller error response during static home posture

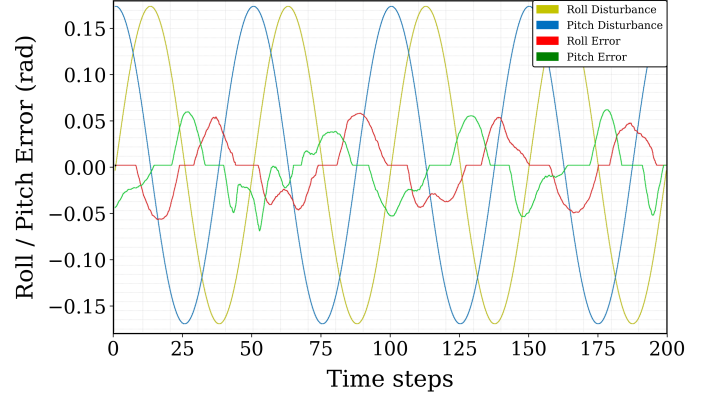


Fig. 6: Controller error response during dynamic trotting gait

error within the rigid 0.08 rad dynamic threshold defined by the proportional gain schedule. This decisive corrective response mitigates external dynamic shifts and effectively prevents long-term drift accumulation, ensuring steady, continuous planar ambulation.

VI. CONCLUSION AND FUTURE WORK

This paper presented a robust posture stabilization and motion generation framework for a custom quadruped robot. By decoupling the rhythmic trajectory planning from the reactive balance control logic into a dual-node architecture, the proposed system achieves fault-tolerant responsiveness and deterministic stability. The implementation of normalized quintic foot trajectories ensures smooth kinematic transitions and minimizes structural impact. Furthermore, integrating

a multistage IMU processing pipeline—featuring dynamic deadband suppression and phase-aware adaptive PID regulation—reliably counters arbitrary external disturbances during both static standing and dynamic locomotion. The quantitative diagnostic validation confirmed that the framework efficiently handles spatial orientation corrections continuously in a real-time ROS 2 environment, successfully bounding positional drift without relying on monolithic or computationally expensive stochastic optimizers.

Future work will focus on the following core areas:

- 1) Advancing state estimation by dynamically fusing localized joint encoder readings with IMU kinematics to track global deformations and deviations.
- 2) Expanding the trajectory generation module to support

variable walking speeds and continuous adaptive stride length modulation for uneven terrain.

- 3) Replacing the current deterministic gain scheduling strategy with optimal control methods, such as Model Predictive Control (MPC), to anticipate and compensate for larger rotational disturbances proactively.

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